University of New Brunswick

Wildfire Risk Prediction

Using Machine Learning

Course: CS 4795

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Abstract:

Wildfires are an increasing global concern due to rising temperatures and changing climate patterns. This project investigates the use of machine learning techniques to predict wildfire risk using environmental and weather-related data. The Algerian Forest Fires dataset from the UCI Machine Learning Repository was used to train and evaluate three classification models: Logistic Regression, Decision Tree, and Random Forest.

The Decision Tree model achieved the best performance, with an accuracy of 93.88% and an F1-score of 0.94. One of the most important findings was that the model produced zero false negatives, meaning that no actual wildfire cases were missed. This is especially critical in real-world applications were failing to detect a fire can have severe consequences. The results demonstrate that machine learning can effectively identify wildfire risk patterns and that simpler models can sometimes perform as well as more complex ones.

Introduction:

Wildfires have become a major environmental and economic issue in many parts of the world. As global temperatures rise and weather patterns become more extreme, the frequency and intensity of wildfires continue to increase. This makes early prediction and prevention more important than ever.

The goal of this project was to explore how artificial intelligence, specifically machine learning, can be applied to predict wildfire risk based on environmental conditions. I chose this topic because it represents a practical and meaningful use of AI, rather than a purely theoretical problem. According to Russell and Norvig (2021), artificial intelligence systems are most

impactful when they can analyze complex real-world data and make informed decisions based on patterns.

Throughout this project, I learned how to work with real-world datasets, preprocess data, implement multiple machine learning models, and evaluate their performance using appropriate metrics. I also gained insight into how different models behave and how increased complexity does not always guarantee improved performance.

The Problem (Explained for a 7th-Grade Student)

Imagine you are trying to predict whether a forest will catch fire based on the weather.

If the day is very hot, dry, and windy, there is a higher chance of a fire starting. If it is cool and rainy, the chance is much lower.

Now imagine teaching a computer to make that same decision. You give it examples of past days, including:

- the weather conditions
- whether a fire happened or not

The problem is to train the computer so that when it sees new weather conditions, it can say:

- "There will be a fire"

  or

- "There will not be a fire"

Key Idea (Explained to a Friend)

Instead of writing rules manually, like:

- “If temperature > 30 and humidity < 20 → fire”

we let the model learn patterns from real data.

This is called supervised learning, where the model is trained using labeled examples. According to James et al. (2021), supervised learning allows models to identify relationships between input variables and outputs by learning from historical data.

In this project, the model looks at environmental features such as temperature, humidity, wind speed, and fire weather index values, and learns what combinations of these conditions are likely to lead to a fire.

Once trained, the model can take new data and make predictions about wildfire risk.

The Details: ***Dataset and Features***

The dataset used in this project is the Algerian Forest Fires dataset from the UCI Machine Learning Repository. It contains 244 instances with no missing values, making it well-suited for machine learning applications.

The features include:

- Temperature
- Relative Humidity
- Wind Speed
- Rain
- FFMC (Fine Fuel Moisture Code)
- DMC (Duff Moisture Code)
- DC (Drought Code)

- ISI (Initial Spread Index)
- BUI (Build Up Index)
- FWI (Fire Weather Index)

These fire weather indices are widely used in wildfire monitoring systems and represent the dryness and flammability of vegetation.

Model Implementation:

The implementation followed a standard machine learning pipeline:

1. Load dataset using the ucimlrepo library
2. Clean and preprocess the data
3. Apply scaling and encoding using a pipeline
4. Split the dataset into training (80%) and testing (20%)
5. Train three models:
    - Logistic Regression
    - Decision Tree
    - Random Forest
6. Evaluate performance using accuracy, F1-score, and confusion matrix
7. Save the best-performing model

The models were implemented using the scikit-learn library (Pedregosa et al., 2011), which provides efficient tools for building and evaluating machine learning models.

Overall Performance:

The performance of the models was as follows:

- Logistic Regression: 91.84% accuracy, F1-score 0.92
- Decision Tree: 93.88% accuracy, F1-score 0.94
- Random Forest: 93.88% accuracy, F1-score 0.94

Although Random Forest is generally expected to outperform a single Decision Tree (Breiman, 2001), both models achieved identical performance in this case.

Most Striking Success:

The most important success was:

❖ **Zero false negatives**

This means that the model correctly identified all actual fire cases and did not miss any. In real-world wildfire prediction systems, missing a fire is far more dangerous than raising a false alarm. Therefore, this result demonstrates that the model is highly reliable for safety-critical applications.

Most Striking Failure

The main limitation was: ***Three false positives***

These are cases where the model predicted a fire when there was none. While this could lead to unnecessary warnings or resource allocation, it is generally considered a safer error compared to missing real fires.

The confusion matrix visualization clearly shows this trade-off between safety and precision.

Related Works:

There are several alternative machine learning approaches that could be applied to wildfire prediction.

For example:

- Neural Networks, which can model complex nonlinear relationships (Goodfellow et al., 2016)
- Support Vector Machines, which are effective in high-dimensional spaces
- Gradient Boosting methods such as XGBoost, which often achieve state-of-the-art performance

However, these methods were not chosen for this project because:

- They require more tuning and computational resources
- They are less interpretable than simpler models
- The dataset used in this project is relatively small and structured

According to Bishop (2006), model selection should consider both performance and interpretability. In this case, Decision Trees provided a strong balance between accuracy and simplicity.

Conclusion:

This project successfully demonstrated the use of machine learning to predict wildfire risk using environmental data. The Decision Tree model achieved the best performance, with 93.88% accuracy and zero missed fire cases. This highlights the effectiveness of machine learning in identifying patterns in wildfire-related conditions.

A key takeaway from this project is that increased model complexity does not always lead to better results. Despite Random Forest being a more advanced method, it did not outperform the simpler Decision Tree model on this dataset.

Contribution:

This project provides a practical example of applying machine learning to a real-world environmental problem. It demonstrates the full pipeline from data preprocessing to model evaluation and can serve as a useful reference for students learning about AI and predictive modeling.

Future Work:

Future improvements could include:

- Hyperparameter tuning to reduce false positives
- Using larger and more diverse datasets (e.g., Canadian wildfire data)
- Performing feature importance analysis
- Testing advanced models such as XGBoost
- Developing a real-time prediction system

Bibliography:

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Appendix:

Source code:

The complete implementation of this project, including training, evaluation, and visualization scripts, is available in the project's GitHub repository. The repository contains all Python files used to build and test the models, along with generated outputs such as performance metrics, confusion matrices, and comparison graphs.

Failed Dataset:

The project used a single well-structured dataset (Algerian Forest Fires dataset), which contained no missing values and required minimal preprocessing. Because of the quality and completeness of this dataset, no alternative or failed datasets were needed during the development process.